

No. of Printed Pages : 4
Roll No.

188451

Level-5, 1st Sem/ DVOC (Ref. & Air Cond.)
Subject : Basics of Refrigeration

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple choice questions. All questions are compulsory (5x1=5)

- Q.1 Refrigeration is based on
- a) First law of Thermodynamics
 - b) kelvin Plank's statement of 2nd law
 - c) Clausius statement of 2nd law of thermodynamics
 - d) 3rd law of Thermodynamics
- Q.2 Three Ton Refrigeration is equivalent to
- a) 5KJ/s
 - b) 10KJ/s
 - c) 10.5Kw
 - d) 7KW
- Q.3 In which part cooling is achieved in refrigerator?
- a) Evaporator
 - b) Capillary
 - c) Condenser
 - d) Compressor
- Q.4 Which part is not essential in absorption refrigeration system?
- a) Absorber
 - b) Condenser
 - c) Generator
 - d) Compressor

Q.5 What is chemical formula of Ammonia?

- a) CO_2 b) NH_4
c) CH_3 d) CUSO_4

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

- Q.6 Refrigeration refers to process of _____.
Q.7 Chemical formula of Carbon-dl-oxide is _____.
Q.8 Molecular mass of ammonia is _____.
Q.9 Carnot cycle is a cycle of _____ efficiency.
Q.10 Effect of _____ is to increase the efficiency of refrigeration cycle.

SECTION-C

Note: Very Short answer type questions. Attempt any six questions out of eight questions. (6x5=30)

- Q.11 Explain in all unit 1 TR.
Q.12 Explain the working of condenser.
Q.13 Give 5 properties of CO_2 as a refrigerant?
Q.14 Give any 5 refrigerants and their applications.
Q.15 State difference between NH_3 & CO_2 .

Q.16 Explain components of Carnot Cycle.

Q.17 Compare the merits and demerits of wet and dry compression.

Q.18 Classify refrigerants with examples.

SECTION-D

Note: Long answer type questions. Attempt any one question out of two questions. (1x10=10)

- Q.19 Explain Vapour Compression Refrigeration System in detail with diagram. Also draw its PV & TS diagram.
Q.20 Explain the effect of sub cooling & superheating on COP of cycle with a diagram.